

# Turbulence-Resilient Self-Aligned Bidirectional Optical Wireless Using Optical Phase Conjugation

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**Abstract** We demonstrate a bi-directional OWC link in which an optical phase-conjugate uplink provides automatic alignment and turbulence compensation, enabling a reverse high-speed downlink that achieves a 10.1-Gb/s peak throughput while sustaining Gb/s-class transmission over an optical power equivalent distance of 80 m. ©2026 The Author(s)

## Introduction

Underwater wireless optical communication (UWOC) is gaining increasing traction as a high-capacity alternative to conventional acoustic links, especially for short-range applications. Emerging applications such as untethered inspection and underwater robotics demand communication links with high throughput and low latency. Optical carriers in the low-absorption blue-green window can deliver multi-Gb/s data rates over tens of meters [1]. This capability enables real-time control and dense sensor backhaul in situations where deploying cables is impractical.

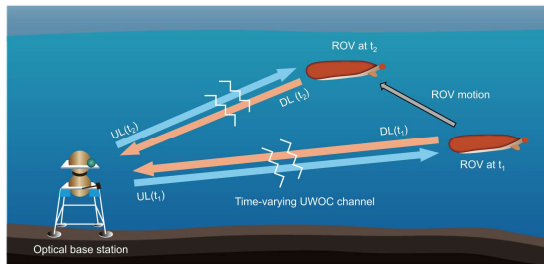


Fig. 1: Scheme of underwater remote-controlled ROV.

Despite these advantages, practical UWOC systems still face challenges in dynamic environments due to signal misalignment and severe wave-front distortion [2]. Figure 1 shows a representative communication scenario involving a rapidly moving remotely operated vehicle (ROV) with UWOC capability. Narrow beam transmission systems require pointing acquisition and tracking to allow communications. Existing approaches are broadly divided into active and passive schemes. Active methods, such as beacon-assisted tracking and mechanical scanning, offer high accuracy but incur substantial complexity, cost, and power consumption [3,4]. Passive methods, including wide field-of-view reception, retroreflectors, and detector arrays, reduce system complexity but are often limited in flexibility and power efficiency [5,6].

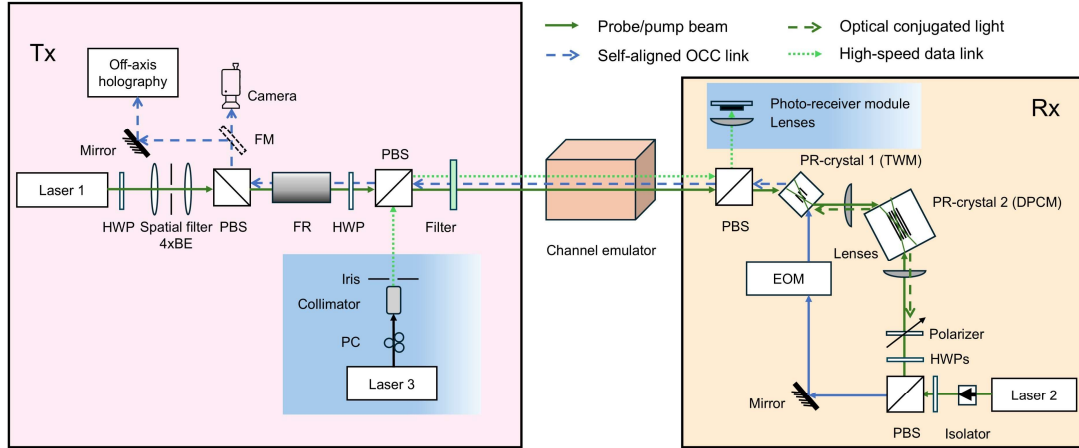
Apart from alignment, links also suffer from wavefront distortion caused by underwater turbulence effects. Progress in adaptive optics has demonstrated several underwater turbulence mitigation schemes, but power demand and

computational complexity limit achievable range and latency [4,7]. While detector-array diversity can boost collected signal power, it suffers from limited bandwidth and faces challenges in practical deployment due to strict form factor requirements [8]. Other approaches, such as iterative digital signal processing (DSP) stacks or photonic integrated circuits, depend on tightly tuned designs and transfer poorly to the highly variable conditions of underwater channels [9,10]. A practical solution should mitigate distortion without relying on complicated DSP, support robust automatic alignment, maintain compactness, and operate efficiently on low power for field deployment.

In this paper, we propose a novel bidirectional optical wireless architecture using optical phase conjugation (OPC). Within its operating field of view, the uplink can be automatically established. The proposed OPC-link can track and compensate turbulence-induced channel fluctuations with dominant temporal frequencies up to ~0.5 Hz, which covers weak dynamic-turbulence regime [11]. As the conjugated wave retraces its path, time-reversal reciprocity automatically corrects accumulated wavefront errors, partially restoring the signal after propagation through turbulence. Another high-speed 520-nm downlink is also established by polarization multiplexing. This link can achieve a peak throughput of 10.1 Gb/s and sustains 1 Gb/s over an optical power equivalent underwater distance of 80 m.

## Principle and Methodology

A dual-phase conjugate mirror (DPCM) is an optical phase-conjugation system that generates a time-reversed replica of an incident wavefront to compensate for distortions. The typical structure of DPCM can be found in [12-15]. In this experiment, a hybrid receiver based on DPCM and two-wave mixing (TWM) is employed to balance the trade-off between phase-conjugate build-up time and amplification gain. Here, the response time of the DPCM-TWM receiver is defined as the sum of the build-up times of the DPCM and TWM processes. For oceanic mesoscale turbulence ( $10^{-2}$ ~ $10^1$ Hz), if the response is sufficiently



**Fig. 2:** The bi-directional optical wireless system configuration. FM: flip mirror; PC: polarization controller.

fast relative to channel variations, the channel appears quasi-static, enabling effective wavefront reconstruction by the OPC.

An underwater channel emulator is employed in the in-lab demonstration to simulate light transmission over various underwater distances, including attenuators, a beam expander and a spatial light modulator (SLM). The average power density of the acceptance surface  $\bar{E}_{(z)}$  can be calculated by

$$\bar{E}_{(z)} = P_t e^{-cz} \cos^2 \theta_h / \pi z^2, \quad (1)$$

where  $P_t$  is the emitted power at the transmitter,  $c$  is the equivalent attenuation coefficient accounting for absorption and scattering in water,  $z$  is the transmission distance, and  $\theta_h$  is the divergence half-angle. We consider a  $c = 0.15\text{m}^{-1}$ , which is a conservative value for moderately clear water [16].

### System Operation

Figure 2 shows the bi-directional system configuration. The system comprises an optical camera communication uplink based on OPC and a downlink based on polarization multiplexing.

**Self-aligned uplink.** A 532-nm DPSS laser (Laser 1) at the transmitter provides the probe beam. Its polarization is set to p-polarization by a half-wave plate (HWP), and the beam is spatially filtered and expanded to 7.4 mm to reduce divergence. A polarizing beamsplitter (PBS) and a Faraday rotator (FR) form an optical isolator, separating the transmit and return paths. The probe is then polarization-multiplexed with the high-speed signal and transmitted through a bandpass filter into the channel.

At the receiver, the probe passes through a PBS and a photorefractive crystal (PR1) and is then focused into a second crystal (PR2) for double phase conjugation. A second 532 nm laser (Laser 2) provides a counter-propagating reference beam in PR2 to record the wavefront, with equal irradiance to the probe at the input surface

of the crystal. This process generates a phase-conjugated beam that propagates back along the incident path.

The conjugated beam is further amplified in PR1 via TWM using a local pump beam from Laser 2. The pump irradiance is set significantly higher than that of the conjugated beam to ensure sufficient gain. Intensity modulation is applied to the pump using an external optical modulator (EOM) and transferred to the amplified conjugated beam, which preserves both the wavefront information and the modulation.

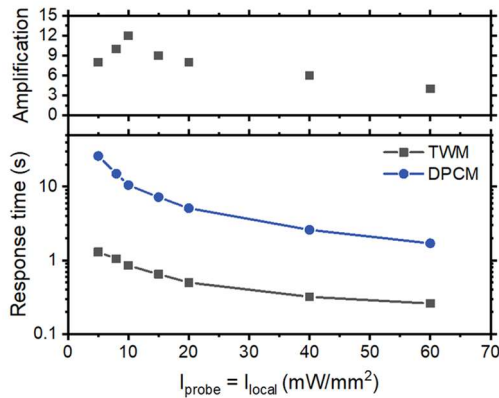
The returned signal is separated by the isolator and detected by a high-speed camera. The bandpass filter provides angular selectivity, suppressing background noise while maintaining signal quality. Direct intensity demodulation at the camera enables a robust, low-rate control uplink. Off-axis holography is used to analysis wavefronts of the received beams.

**High-speed Downlink.** A 520-nm laser diode generates the high-speed signal, which is modulated and collimated to match the probe beam diameter. The signal is s-polarized and multiplexed with the probe at the transmitter via a PBS. It co-propagates with the probe along the same path to the receiver, where it is separated by a PBS and directed to photoreceiver modules for detection. The received signal is then demodulated off-line to recover the transmitted data. For short- and long-range links, a photodetector (Thorlabs DET025A) and a silicon photomultiplier (SiPM, Onsemi J30020) are employed, respectively, to optimise data rate and ensure stable transmission.

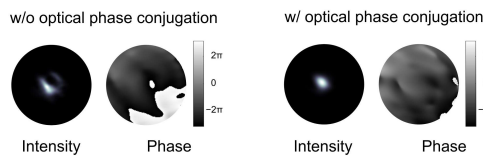
In this configuration, a bidirectional link is established. The high-throughput downlink operates independently, while the self-aligned uplink loop is maintained to support alignment and dynamic distortion compensation.

### Result and Discussion

Figure 3 shows the DPCM and TWM build-up times under varying probe/local light intensity. In



**Fig. 3:** DPCM and TWM build-up times and amplification factor under different power density.

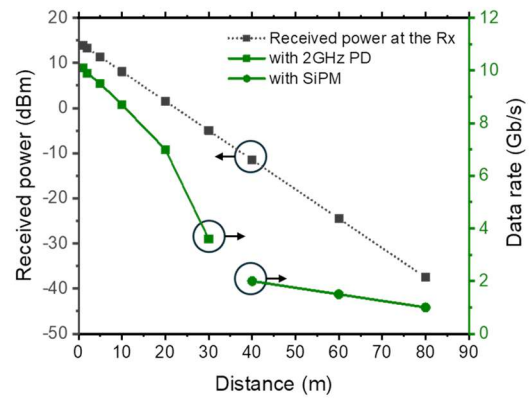


**Fig. 4:** Beam profiles of received uplink beam

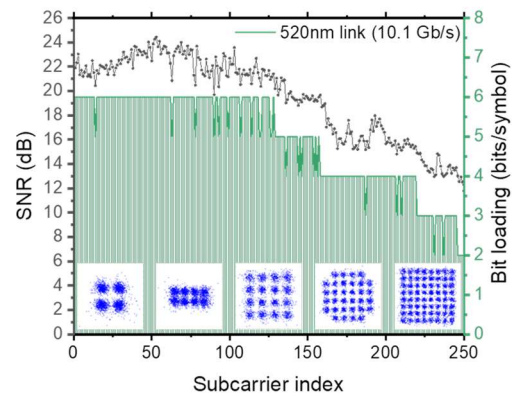
the TWM stage, the phase-conjugate wave denotes the phase-conjugated beam generated by DPCM in PR crystal 2 and injected into PR crystal 1 for amplification. As the input intensity increases from 5 mW/mm<sup>2</sup>, the build-up times of both DPCM and TWM decrease rapidly at first, then gradually saturate due to high-power injection. At 60 mW/mm<sup>2</sup>, the DPCM-TWM response time is reduced to below 2 s. The gain initially increases sharply with probe intensity, reaches a maximum gain of 12 around 10 mW/mm<sup>2</sup>, and then decreases as saturation effects dominate. Unlike passive retroreflectors, the phase-conjugated return is actively amplified, enabling compensation of propagation loss and improved resilience to channel impairments.

Based on the results shown in Fig. 3, an input intensity of 20 mW/mm<sup>2</sup> is selected as the optimal operating point for further evaluation. At this condition, the combined DPCM-TWM response time is 5.5 s. Slowly varying phase distortions are emulated using computer-generated holograms programmed on the SLM in the underwater channel emulator. Off-axis holography is employed to measure the intensity and phase distributions with and without phase conjugation, as shown in Fig. 4. With the DPCM-TWM receiver, the returned wavefront is effectively restored (>90%), and the signal intensity is enhanced due to TWM amplification. This enables both self-alignment and an increased link power budget.

Figure 5 presents the received optical power and achievable data rate versus power equivalent distance for the downlink. Within a 30m emulated underwater range, the optical power budget of the point-to-point link remains sufficient.



**Fig. 5:** Received optical power and data rate versus equivalent distance for the downlink.



**Fig. 6:** Measured SNR and bit-loading profile with constellation diagrams.

Therefore, a bit-loading with power-allocation scheme was applied to maximize the achievable communication rate. The maximum rate can reach 10.1 Gb/s at a distance of 1-m. The corresponding signal-to-noise ratio (SNR) profile, per-subcarrier bit allocation, and constellation diagrams at this rate are shown in Fig. 6. As the range increases, the received optical power drops rapidly. The photoreceiver module is changed from a Si detector to a high-sensitivity SiPM, and a non-return to zero on-off keying scheme is adopted for power-limited operation. The results show that as the emulated transmission distance increases from 40 to 80 m, the link can still maintain a data rate of 1Gb/s.

## Conclusion

This study presents an OPC-enabled turbulence-resilient self-aligned bidirectional optical wireless system designed for underwater applications. By employing a DPCM-TWM receiver, the system achieves a rapid response time of up to 2 s and an optical gain of 12× over an emulated channel. A high-speed downlink employing bit loading and power allocation achieves a peak data rate of 10.1 Gb/s at 1 m and sustains 1 Gb/s over an optical power equivalent distance of 80 m.

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